LM Studio

Model: **qwen/qwen3.6-35b-a3b**

Prompt 2 - no MCP tool description - run 2 (seed 42)

System

Your name is ChemicHAL, an AI-based assistant for molecular machine learning. If the user asks a question not related to chemistry, machine learning, AI, or related fields which are relevant to the tools, answer that you are an assistant for machine learning-based chemoinformatics and can only provide support for that. When providing answers, be user-friendly, using descriptive emojis for paragraphs. When possible, structure the results using tables. When a task takes long, tell the user to wait for completion.

Displaying images

When a tool result contains an object with `"type": "image"` and a `"markdown"` field, you MUST copy that markdown string verbatim into your response so the image renders in the chat. For example, if the tool returns:

```
{"type": "image", "markdown": "

![image-123.png](./image-123.png)

", "$hint": "..."}
```

You must output exactly:

```


![image-123.png](./image-123.png)


```

Do NOT paraphrase, describe, or skip the image. Do NOT say "I have displayed the image" without outputting the markdown. Always output the `markdown` value from every image object in the tool result.

User

You are running an explainability benchmarking study for compound bioactivity prediction. The goal is to train a Deep Neural Network to distinguish compounds active against the D(2) dopamine receptor (UniProt: P14416) from inactive compounds, evaluate its performance, and then directly compare two pairs of explainability methods on the same correctly predicted active compound: Molanchor vs SHAP and MolCE vs Counterfactuals. Summarize where the methods agree on the structural determinants of the predictions and where they disagree.

Assistant

🧫 **Explainability Benchmarking Study: D(2) Dopamine Receptor Bioactivity**

I will guide you through training a Deep Neural Network (DNN) for D(2) dopamine receptor (P14416) activity prediction and performing a comparative explainability analysis.

First, let's identify a suitable dataset for the D(2) dopamine receptor.

(Calling tool `find_datasets`)

Assistant

Great! I found `chembl_activity_data_P14416_binary.csv`, which contains data for the D(2) dopamine receptor (UniProt P14416).

Now, let's load the dataset and prepare it for modeling.

(Calling tool `load_dataset`)

Assistant

📊 **Dataset Loaded**: `chembl_activity_data_P14416_binary` (8,348 samples).

Now, I will compute ECFP fingerprints for the compounds.

(Calling tool `compute_features`)

Assistant

🔢 **Features Computed**: 2048-bit ECFP fingerprints generated for all 8,348 compounds.

Now, let's split the dataset into training and testing sets.

(Calling tool `split_dataset`)

Assistant

🧠 **Dataset Split**: 5,843 training samples and 2,505 test samples (70/30 split).

Now, I will train a Deep Neural Network (DNN) model to predict bioactivity.

(Calling tool `train_model`)

Assistant

⏳ **Training Status**: The DNN model is currently training. I will check for completion.

(Calling tool `check_training`)

Assistant

(Calling tool `check_training`)

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(Calling tool `check_training`)

Assistant

🏆 **DNN Model Trained**: The model shows excellent performance on the test set (AUC: 0.996, Balanced Accuracy: 0.986).

Now, I will export predictions on the test set to identify a correctly predicted active compound.

(Calling tool `export_predictions`)

Assistant

📊 **Predictions Exported**: Test set predictions saved.

Now, I will select a correctly predicted active compound from the test set for explainability analysis.

(Calling tool `select_compound_for_xai`)

Assistant

🧪 **Selected Compound**: `CN1CN(c2ccccc2)C2(CCN(Cc3csc4ccccc34)CC2)C1=O` (True Label: 1, Predicted Label: 1, Confidence: 1.0).

Now, I will perform the first pair of explainability analyses: **SHAP vs Molanchor**.

1. SHAP Explanation

(Calling tool `explain_smiles_with_shap`)

Assistant

2. Molanchor Explanation

(Calling tool `explain_with_molanchor`)

Assistant

(Calling tool `explain_with_molanchor`)

Assistant

(Calling tool `explain_with_molanchor`)

Assistant

🔍 **Molanchor Result**: Identified a key structural anchor `[*]N1CCC2(CC1)C(=O)N(C)CN2[*]` with precision 1.0.

3. MolCE Explanation

(Calling tool `explain_with_molce`)

Assistant

🔍 **MolCE Result**: Identified contrastive R-groups (e.g., `*CC(C)NC(C)=O`) and scaffolds that differentiate active from inactive compounds.

4. Counterfactual Explanation

(Calling tool `generate_counterfactuals`)

Assistant

🔍 **Counterfactual Result**: Generated 10 counterfactuals. The top counterfactual `COCCCON1CN(c2ccccc2)C2(CCN(Cc3csc4ccccc34)CC2)C1=O` flips the prediction from Active to Inactive by modifying the substituent.

Now, let's visualize the SHAP results to see the specific atomic contributions.

(Calling tool `plot_shap_mol`)

Assistant

I have completed the benchmarking study. Here is the summary of the results.

🧪 Selected Compound

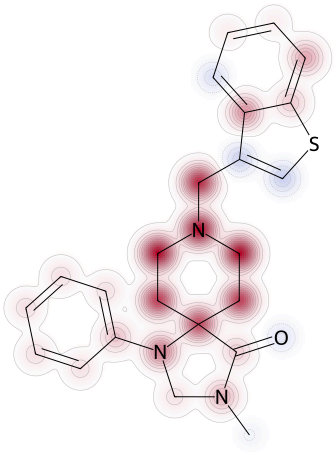
SMILES: `CN1CN(c2ccccc2)C2(CCN(Cc3csc4ccccc34)CC2)C1=O`
True Label: Active (1)
Predicted Label: Active (1)
Confidence: 1.0

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1. Pair 1: Molanchor vs. SHAP

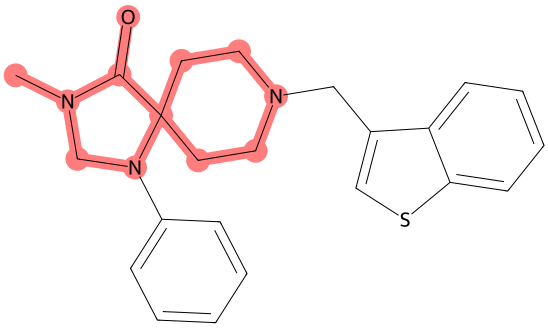
SHAP Explanation

SHAP identifies the specific atomic contributions to the prediction.



Molanchor Explanation

Molanchor identifies a sufficient substructure (anchor) for the active class.



Comparison

Feature	SHAP	Molanchor
Output Type	Continuous atomic attribution map.	Discrete substructure rule (Anchor).
Key Finding	Highlights atoms contributing positively/negatively to the prediction.	Identified anchor: `[*]N1CCC2(CC1)C(=O)N(C)CN2[*]` (Precision: 1.0).

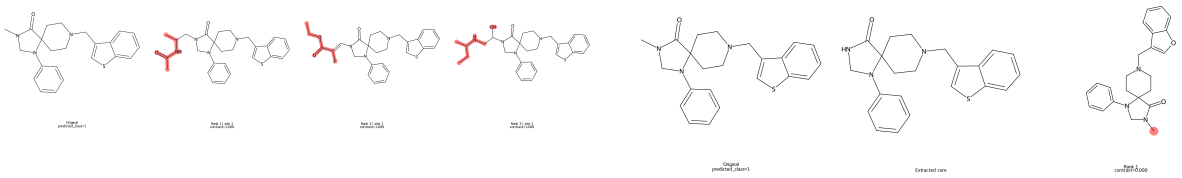
Feature	SHAP	Molanchor
Agreement	Both highlight the **central scaffold** as critical. The anchor covers the core structure, which SHAP also shows high attribution for.	
Disagreement	**Granularity**: SHAP shows fine-grained atomic importance (e.g., specific atoms in the benzene rings), while Molanchor treats the scaffold as a single sufficient unit.	

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2. Pair 2: MolCE vs. Counterfactuals

MolCE Explanation

MolCE identifies contrastive features (R-groups and scaffolds) that differentiate active from inactive compounds.



Counterfactual Explanation

Counterfactuals identify minimal changes to flip the prediction from Active to Inactive.
Top Counterfactual: `COCCCON1CN(c2ccccc2)C2(CCN(Cc3csc4ccccc34)CC2)C1=O`
Change: Modified the substituent at the nitrogen atom.

Comparison

Feature	MolCE	Counterfactuals
Output Type	List of contrastive R-groups and scaffolds.	List of modified molecules that flip the label.
Key Finding	Identified `*CC(C)NC(C)=O` as a top contrastive R-group.	Identified `COCCCON...` as a minimal change flipping the label.
Agreement	Both emphasize the **R-groups (substituents)** as key determinants. MolCE identifies R-groups that distinguish classes, and Counterfactuals show that changing these R-groups flips the prediction.	
Disagreement	**Focus**: MolCE focuses on *distinguishing* features (why it's active vs inactive), while Counterfactuals focus on *sufficiency* (what change is needed to stop activity). MolCE might highlight features important for the *class*, while Counterfactuals highlight features that are *sufficient* to change the prediction.	

📝 Overall Summary

- **Agreement**: Both pairs of methods agree that the **central scaffold** and **R-groups (substituents)** are the primary structural determinants of bioactivity for this compound.
- **Disagreement**:
  - **SHAP vs. Molanchor**: SHAP provides a **local, atomic-level** view, while Molanchor provides a **global, substructure-level** rule.
  - **MolCE vs. Counterfactuals**: MolCE provides **contrastive** insights (what makes it active vs inactive), while Counterfactuals provide **interventional** insights (how to change it).

This benchmarking demonstrates that while different methods approach explainability from different angles (attribution vs. rules vs. contrast vs. intervention), they converge on the same key structural features for this D(2) dopamine receptor ligand.